Venkata Lakshmi Parimala Pasupuleti

Staff Software Engineer · PropTech & Apartment AI · End-to-End Product Ownership · Production AI Systems · Scalable Architecture · Customer-Facing Delivery · Full Autonomy

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PROFESSIONAL SUMMARY

Software engineer with 8 years of end-to-end product ownership in apartment and affordable housing software - the exact domain EliseAI operates in. At Entrata, I didn't follow a playbook; I built one. Owned the HUD income-limit compliance pipeline for 7 years (federal data → 180,000 domain records powering eligibility decisions for 100K+ housing units), designed scalable architecture for async processing, and shipped 20+ production-grade features across leasing, compliance, and document workflows - all customer-facing, all owned solo from design through deployment. Two AI systems extend this pattern: LeaseI (3-surface agentic AI platform for apartment compliance, built concept-to-production with full testing, observability, and CI integration) and AgentProbe (open-source LLM evaluation platform with CI-integrated regression gate - zero regressions across 1,240+ test cases). I move fast, take full ownership, and thrive in the startup environment where the playbook is yours to write. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Product Engineering & Scalable Architecture: Python (8yr), System Design, Scalable Architecture, End-to-End Product Ownership, FastAPI, REST APIs, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ, Docker, AWS (S3, EC2, RDS, Glue), CI/CD (GitHub Actions), Microservices, Distributed Systems, Multi-Tenant SaaS, Linux, Git

AI & Autonomous Systems: Claude Code, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Tool Calling, LLM Evaluation, LLM-as-Judge, Hallucination Detection, RAG Pipelines, ChromaDB, Embeddings, Prompt Engineering, Anthropic API, OpenAI API, MCP, HITL, Guardrails, OCR, spaCy, BERT, NLP, PyTorch, scikit-learn

Reliability & Delivery: Testing Strategy (Unit, Integration, End-to-End), Observability & Monitoring, Performance Engineering, Production Debugging, Continuous Integration, Deployment Automation, PHI Compliance, Data Governance, Code Reviews & Mentoring, Cross-Functional Stakeholder Delivery, Agile/Scrum, TypeScript, JavaScript, React.js

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - End-to-End Delivery

- Scoped and shipped a production-grade agentic AI system (LangGraph, FastAPI, Python) from ambiguous customer requirements to deployment - used Claude Code throughout to compress the build cycle; delivered working software in weeks; owned architecture, testing, observability, and stakeholder handoff end-to-end with full autonomy.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and system design.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **End-to-end ownership of production-grade systems for apartment operators** - 20+ customer-facing production modules that property managers, leasing agents, and compliance teams relied on daily; owned HUD income-limit pipeline for 7 years (federal source → 180,000 domain records powering eligibility and rent calculations for 100K+ housing units); designed scalable architecture from scratch and iterated on it in production across 120+ Agile sprints - no playbook, just ownership.
- **Scalable, reliable infrastructure built to operate autonomously** - RabbitMQ async processing with retry logic, idempotency guarantees, and dead-letter handling eliminating double-processing failures; Docker containerization (50% scalability improvement, 30% deployment time reduction); AWS S3/EC2/RDS/Glue; Redis caching; 200+ SQL query optimizations (30% performance gain); 99.5% uptime across multi-tenant production environments; 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents 7yr).
- **Testing, observability, and CI - built into every delivery** - GitHub Actions CI/CD; maintained sub-10 bug count for 6+ consecutive months; built internal debugging tools reducing triage time by 40%; 300+ peer code reviews enforcing production-grade standards; leveraged Claude Code and AI-assisted development to accelerate iteration and ship faster; mentored 5+ engineers.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for apartment management and affordable housing compliance; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: Agentic AI Platform for Apartment Compliance - Concept to Production

PropTech · Autonomous Systems · End-to-End

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface production-grade AI platform built solo from discovery through deployment - no playbook, full ownership: document intelligence (OCR, section-aware chunking, 100% citation correctness), 8-agent LangGraph support investigation (autonomous tool routing, memory, HITL escalation), and tenant self-service with deterministic guardrails. Full testing strategy, observability, and CI integration - 93.6% precision, 87.9% F1 across 1,240 apartment compliance test cases.

AgentProbe: LLM Evaluation with CI-Integrated Regression Gate

Observability · Testing · Open Source

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform built end-to-end: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking regressions before they ship, multi-provider routing for cost optimization. Zero regressions shipped across 1,240+ test cases. Open-source.

AI on FHIR: Full-Stack NLP System - Customer-Facing Production

Full-Stack · APIs · Customer-Facing

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack customer-facing AI application (Python backend + React.js/TypeScript frontend) translating natural-language queries into FHIR REST API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, and full data isolation.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017